

MoPay

MTN & Orange Android Application User Manual

Customer • Merchant • Agent
C2C • C2M • C2A • A2C • M2M • NFC



Professional operating guide for the MoPay multi-rail wallet application concept.

MTN edition

Orange edition

Note: interface visuals in this manual are polished interface previews based on the current application design. Operator branding/API authorization remains subject to MTN or Orange approval.

1. Product Overview

MoPay is structured as a secure digital-wallet application with separate Customer, Merchant and Agent experiences. The same internal ledger supports multiple transaction channels, while MTN MoMo and Orange Money are treated as external operator rails.

Role	Primary functions
Customer	Balance, Top Up, Send, Pay, Receive, NFC card, PIN, freeze/unfreeze, history
Merchant	NFC acceptance, C2M, M2M, daily activity, reports, wallet operations
Agent	A2C, C2A, wallet, history, operator integration status

Supported transaction channels

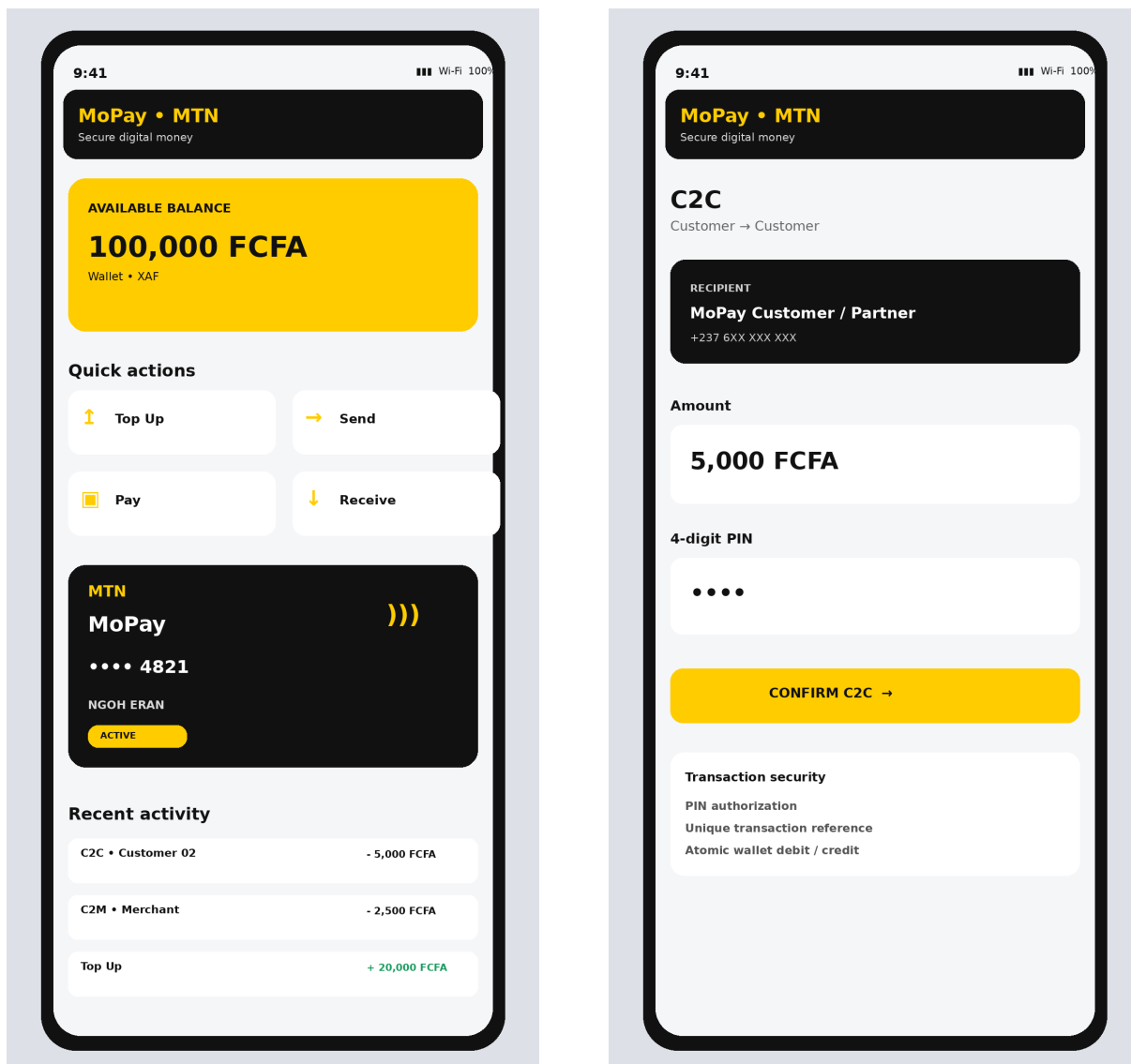
C2C Customer → Customer **C2M** Customer → Merchant **C2A** Customer → Agent

A2C Agent → Customer **M2M** Merchant → Merchant **NFC** Contactless card payment

The transaction engine is designed around atomic debit/credit operations so the sender and recipient ledger states remain consistent.

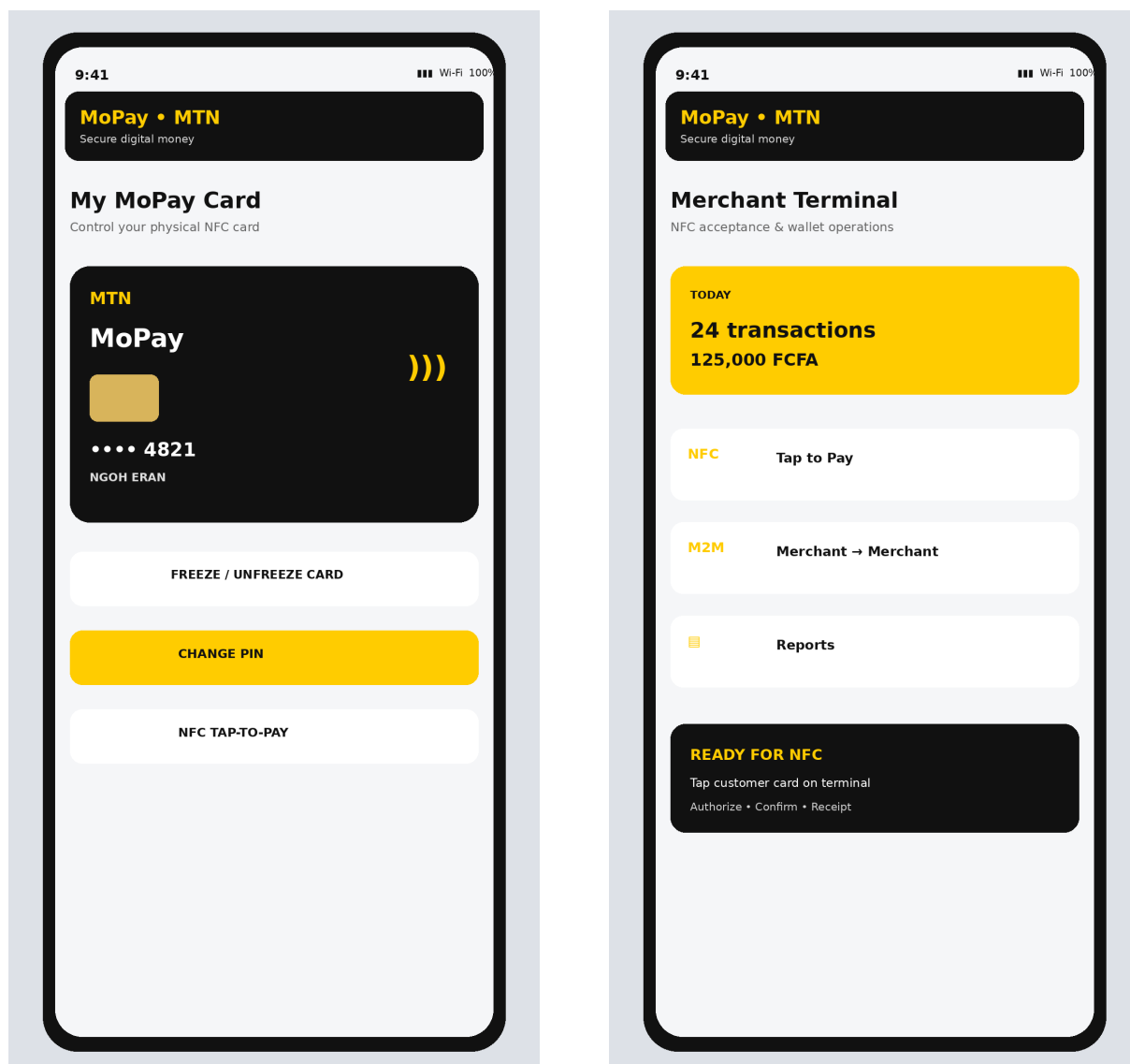
2. MTN Edition

The MTN edition uses a yellow/black visual system inspired by the supplied reference. The card presentation uses a premium contactless-card treatment with masked number, chip, NFC symbol and customer name.



Home: balance, quick actions, card, recent activity. **C2C:** recipient, amount, PIN and confirmation.

3. MTN Card & Merchant Experience



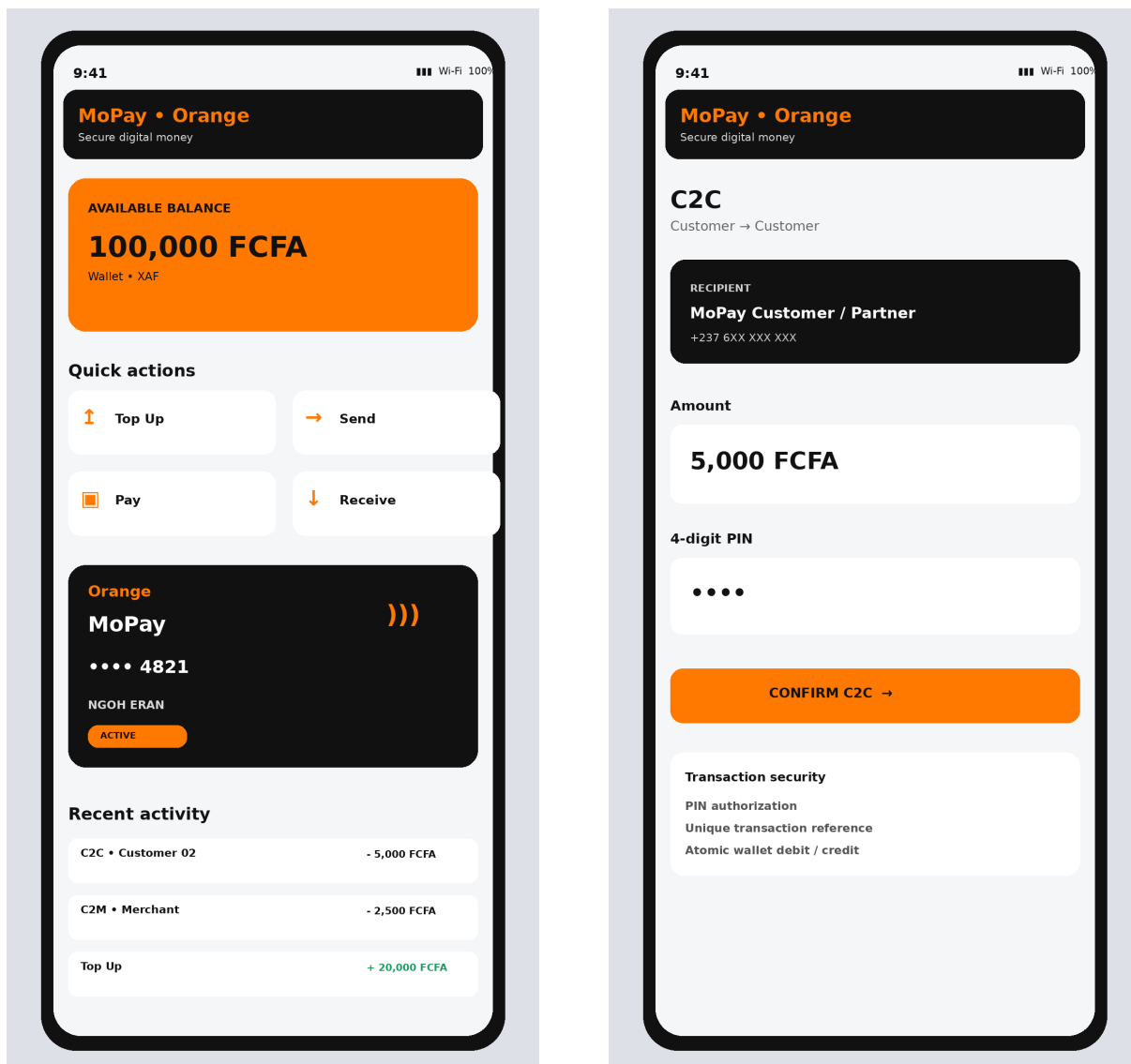
Card management: freeze/unfreeze, change PIN and NFC readiness. **Merchant:** NFC acceptance, M2M and reporting.

NFC payment procedure

1. Merchant enters or confirms the amount. 2. Customer presents the supported NFC card. 3. Merchant terminal detects the card. 4. Transaction is authorized. 5. A success/decline result and reference are recorded.

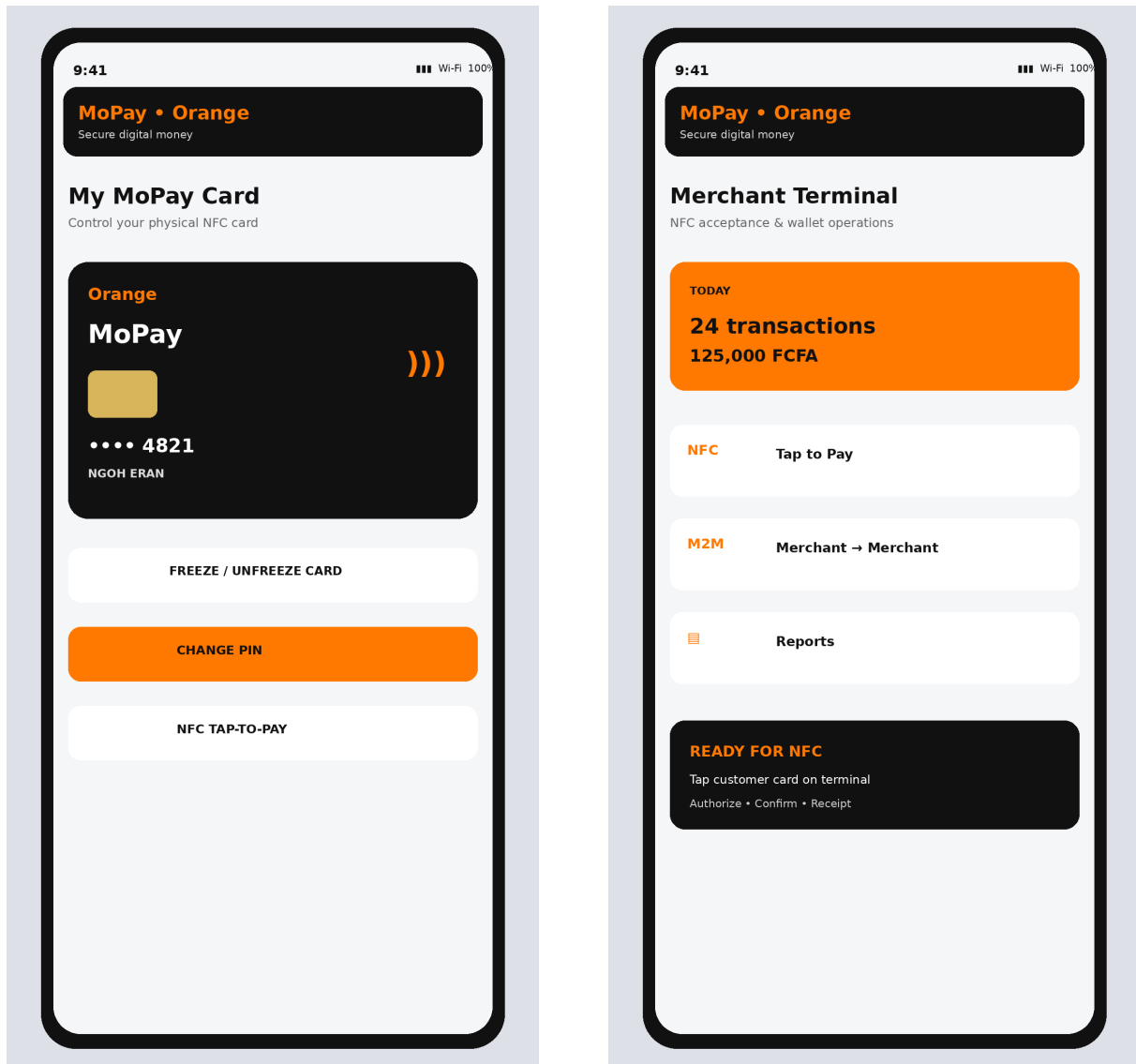
4. Orange Edition

The Orange edition uses an orange/black visual system. The functional structure is shared with the MTN edition while the product presentation, accent colors and card treatment follow the Orange visual direction.



Home: wallet overview and quick actions. **C2C:** secure customer-to-customer transfer.

5. Orange Card & Merchant Experience



Card management: freeze/unfreeze, PIN and NFC readiness. **Merchant:** NFC acceptance, M2M and reports.

6. Customer Operations

Top Up: enter the amount and confirm. In the final operator integration, the funding request is routed through the approved MTN or Orange API rail.

Send: choose C2C, C2M or C2A, select the recipient, enter amount and PIN, then confirm.

Receive: display the customer's MoPay identity/details for an incoming transfer.

Card: view card, freeze/unfreeze and change PIN. A frozen card should be rejected by the payment authorization layer.

History: view transfers, payments, references, status and amounts.

Profile / Settings: manage account information, security and application preferences.

7. Merchant Operations

NFC C2M: the merchant terminal reads the supported payment card and submits the authorization request.

M2M: merchant-to-merchant transfer uses the same secure transaction ledger.

Reports: transaction count, successful/declined activity, amounts and references can be presented for operational review.

8. Agent Operations

A2C: agent sends funds to a customer. **C2A:** customer sends funds to an agent. Agent history records incoming and outgoing activity.

9. Security & Production Architecture

The APK should never contain MTN or Orange secret credentials. The secure production pattern is:

MoPay backend / Supabase Edge Function	Validates identity, transaction rules, limits and authorization
Internal ledger	Atomic debit/credit and transaction record
Operator adapter	MTN MoMo or Orange Money API
Operator	Real external wallet / settlement rail

Before production launch, replace demonstration identities with authenticated users, enforce server-side authorization, configure transaction limits and fraud controls, secure callbacks/webhooks, rotate credentials, and complete the relevant operator/regulatory onboarding.

Operator integration checklist

- API base URLs and environments
- OAuth/client credentials or equivalent authentication
- Transaction initiation and status endpoints
- Callback/webhook specification
- Idempotency/reference requirements
- Settlement and reconciliation rules
- Limits, fees and supported transaction types
- Certification / UAT requirements

10. Quick Troubleshooting

Issue	Action
Card not detected	Confirm NFC is enabled, the terminal supports NFC and the card is presented close to the reader.
Transfer declined	Check account/card status, amount, available balance, recipient and PIN.
History not updated	Refresh the wallet/transaction screen and verify backend connectivity.
Operator transaction pending	Use the operator transaction-status/callback flow; do not duplicate the transaction blindly.
APK cannot connect	Check Supabase URL, publishable key, network access and authenticated session.
API credentials exposed	Immediately revoke/rotate them. Never store operator secrets in the APK.

11. Important Product Status

This manual documents the intended operating experience and current application architecture. The MTN/Orange external money rail is not considered live until the relevant operator provides and authorizes the API integration, credentials, technical specifications and required onboarding/certification.

The internal MoPay transaction flows are designed to cover C2C, C2M, C2A, A2C, M2M and NFC payment workflows. Operator-specific behavior such as fees, limits, reversals, settlement and callback semantics must follow the operator's official specifications.

MoPay

Digital money without digital-device dependency.